

Miscellaneous Topics

National Dairy Development Board (NDDB): National Dairy Development Board, located at Anand, Gujarat, set up in 1965 and declared as a statutory body corporate in 1987 under the NDDB Act, is a premier institution to accelerate the pace of dairy development on cooperative lines in the country.

National Fisheries Development Board (NFDB): National Fisheries Development Board (NFDB) was set up in September, 2006, with its head quarters at Hyderabad to realize the untapped potential of fisheries sector in inland and marine fish capture, culture, processing & marketing of fish, and overall growth of fisheries sector with the application of modern tools of research & development.

Coastal Aquaculture Authority: The Coastal Aquaculture Authority (CAA) was established under the Coastal Aquaculture Act, 2005 vide Gazette Notification dated 22 December 2005. The aims and objectives of the Authority are to regulate 'Coastal Aquaculture' activities in the areas notified by the Central Government as 'coastal areas' and for matters connected therewith or incidental thereto. The Authority is empowered to make regulations for the construction and operation of aquaculture farms in coastal areas, registration of aquaculture farms and hatcheries, to inspect them to ascertain their environmental impact, remove or demolish coastal aquaculture farms which cause pollution, fixing standards for coastal aquaculture inputs in order to facilitate environmentally responsible and socially acceptable coastal aquaculture

Veterinary Council of India (VCI) is a statutory body established under the provision of Indian Veterinary Council Act, 1984. Veterinary Council of India is responsible for regulating veterinary practices as well as for maintaining uniform standards of veterinary education through Minimum Standard of Veterinary Education Regulations in all veterinary institutes across the country.

Marine Products Export Development Authority (MPEDA) was set up by an act of Parliament during 1972. The erstwhile Marine Products Export Promotion Council established by the Government of India in September 1961 was converged in to MPEDA on 24th August 1972. MPEDA is given the mandate to promote the marine products industry with special reference to exports from the country. It is envisaged that this organisation would take all actions to develop and augment the resources required for promoting the exports of "all varieties of fishery products known commercially as shrimp, prawn, lobster, crab, fish, shell-fish, other aquatic animals or plants or part thereof and any other products which the authority may, by notification in the Gazette of India, declare to be marine products for the purposes of (the) Act. **Under Ministry of Commerce and Industry**

Head office: Kochi

Agricultural and Processed Food Products Export Development Authority (APEDA) was established by the Government of India **Under Ministry of Commerce and Industry**

under the Agricultural and Processed Food Products Export Development Authority Act passed by the Parliament in December, 1985.

APEDA is mandated with the responsibility of export promotion and development of the following scheduled products:

- Fruits, Vegetables and their Products.
- Meat and Meat Products.
- Poultry and Poultry Products.
- Dairy Products.
- Confectionery, Biscuits and Bakery Products.
- Honey, Jaggery and Sugar Products.
- Cocoa and its products, chocolates of all kinds.
- Alcoholic and Non-Alcoholic Beverages.
- Cereal and Cereal Products.
- Groundnuts, Peanuts and Walnuts.
- Pickles, Papads and Chutneys.
- Guar Gum.
- Floriculture and Floriculture Products.
- Herbal and Medicinal Plants.

Head office: New Delhi

Agri Exchange is the name given to the Trade portal. An attempt, first of its kind has been endeavored by APEDA, Govt. of India, where online trading is the specialty. In its stride, joint collaboration of UNCTAD and Ministry of Agriculture has given the portal a shape. Globally buyers and sellers in the Agri business world has been given a platform to offer negotiate and transact a deal. Apart from this, it has been loaded with latest statistics pertaining to India and the world.

Reliable data sources like UN Comtrade, FAO and DGCIS has been used. Regular updation related to the statistics is done. Up to date information on the 671 products (approx.) related to APEDA and products does not falls under the purview of APEDA has been hosted. Information on the SPS Measures of different countries, MRLs, Country Profile, Service providers in the Agri export business and Market news has also become an integral part of the portal.

Fish Exchange is the brainchild of MPEDA, Government of India, that shapes up to be that desired trade facilitation portal. Traders require a common forum for discussion, gathering information, viewing trends and knowing the right places to go for their business. Marine product exporters registered under MPEDA will be given access to many digital facilities that were obscure till recently. Exporters can post their available products and offers, receive buying enquiry alerts, check out the latest trade statistics and plan their trade.

Reliable data sources like UN Comtrade, WTO, DGCIS and a few others have been referred to retrieve and display the accurate and most updated data. Traders will get to know the various Sanitary or Phytosanitary regulations, Trade Barriers, MRLs, Import and Export Regulations, Quality Standards directives and all official marine trade related notifications, not only for India, but for as many countries in the globe as can be retrieved.

Sanitary and phytosanitary measures (SPS): The SPS agreement provides a framework of rules to guide WTO members in the development, adoption and enforcement of sanitary (human or animal life or health) and phytosanitary (plant life or health) measures which may affect trade.

MRLs: (Maximum residue limits): The maximum residue limit, is the maximum amount of pesticide residue that is expected to remain on food products when a pesticide is used according to label directions, that will not be a concern to human health.

National Agricultural Cooperative Marketing Federation of India Ltd: (NAFED) was established on the auspicious day of Gandhi Jayanti on 2nd October 1958. NAFED is registered under the Multi State Co-operative Societies Act. NAFED was setup with the object to promote Co-operative marketing of agricultural produce to benefit the farmers. Agricultural farmers are the main members of NAFED, who have the authority to say in the form of members of the General Body in the working of NAFED.

The objectives of the NAFED shall be to organize, promote and develop marketing, processing and storage of agricultural, horticultural and forest produce, distribution of agricultural machinery, implements and other inputs, undertake inter-state, import and export trade, wholesale or retail as the case may be and to act and assist for technical advice in agricultural, production for the promotion and the working of its members, partners, associates and cooperative marketing, processing and supply societies in India.

Head office: New Delhi

FSSAI: Food Safety and Standards Authority of India is an autonomous body established under the Ministry of Health & Family Welfare, Government of India. The FSSAI has been established under the Food Safety and Standards Act, 2006 which is a consolidating statute related to food safety and regulation in India.

Founded in August 2011 and headquarter at New Delhi

National Cooperative Development Corporation (NCDC) was established by an Act of Parliament in 1963 as a statutory Corporation under the Ministry of Agriculture & Farmers Welfare.

Planning, promoting and financing programmes for production, processing, marketing, storage, export and import of agricultural produce, food stuffs, certain other notified commodities e.g. fertilisers, insecticides, agricultural machinery, lac, soap, kerosene oil, textile, rubber etc., supply of consumer goods and collection, processing, marketing, storage and export of minor forest produce through cooperatives, besides income generating stream of activities such as poultry, dairy, fishery, sericulture, handloom etc.

NCDC has been identified as a Nodal Loaning Entity (NLE) for borrowing resources from the Central Sector Scheme Dairy and Livestock and Livestock Processing and Infrastructure Development Fund (DIDF set-up in NABARD and finance the projects submitted by the Eligible End Borrowers (EEBs).

Head office: New Delhi

The Codex Alimentarius Commission (CAC) is an international food standards body established jointly by the Food and Agriculture organization (FAO) and the World Health Organization (WHO) in May 1963 with the objective of protecting consumer's health and ensuring fair practices in food trade. **Head office** Rome, Italy

National Codex Contact Point: *"The Food Safety and Standards Authority of India (FSSAI) under the administration of Ministry of Health and Family Welfare has been designated as the nodal point for liaison with the Codex, known as "National Codex Contact Point of India (NCCP).*

The National Codex Committee has been constituted by the Food Safety and Standards Authority of India, chaired by the Chairperson, FSSAI for keeping liaison with the Codex Alimentarius Commission.

The Indian Council of Agricultural Research (ICAR) is an autonomous organisation under the Department of Agricultural Research and Education (DARE), Ministry of Agriculture and Farmers Welfare, Government of India. Formerly known as Imperial Council of Agricultural Research, it was **established on 16 July 1929** as a registered society under the Societies Registration Act, 1860 in pursuance of the report of the Royal Commission on Agriculture. The ICAR has its headquarters at New Delhi. The Council is the apex body for co-ordinating, guiding and managing research and education in agriculture including horticulture, fisheries and animal sciences in the entire country. With **101 ICAR institutes** and **71 agricultural universities** spread across the country this is one of the largest national agricultural systems in the world.

The National Dairy Research Institute, Karnal was established at Bangalore in 1923 as Imperial Institute of Animal Husbandry and Dairying. In 1936 it was renamed as Imperial Dairy Institute. After the independence of India, in 1947 it was renamed to its current name. In 1955 the institute's headquarters moved to Karnal. The status of deemed to be university was conferred to the institute in 1989.

The National Dairy Research Institute as the premier Dairy Research Institution undertakes research, teaching and extension activities towards dairy development in the country. Being the National Institute, it conducts basic and applied research with the objective to enhance animal productivity and also to develop cost effective technologies for the benefit of the teeming millions. Further, the Institute provides high quality manpower to meet the human resource requirements for the overall dairy development in the country. The Institute also undertakes extension programmes for transferring the know-how from the laboratory to the farmers' fields.

Krishi Vigyan Kendra: The first KVK was established in 1974 at Puducherry. The KVK scheme is 100% financed by Govt. of India and the KVKs are sanctioned to Agricultural Universities, ICAR institutes, related Government Departments and Non-Government Organizations (NGOs) working in Agriculture.

KVK, is an integral part of the National Agricultural Research System (NARS), aims at assessment of location specific technology modules in agriculture and allied enterprises, through technology assessment, refinement and demonstrations.

Uttar Pradesh has the highest number of KVKs.

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As on date 716 KVKs has been established. Reference ICAR website.

National Programme for Organic Production (NPOP):

- NPOP launched during 2001 laid the foundation for systematic development of organic agriculture sector in the country.
- NPOP is being managed by APEDA under the Ministry of Commerce and Industry.
- NPOP is the main stay for quality assurance for organic agriculture products in India operated through 28 accredited certification bodies.
- All types of agriculture, horticulture and non-food crops are grown under organic certification process.
- Livestock, aquaculture, animal feed processing and handling, mushroom production, sea weeds, aquatic plants and green house crop production have also come under the ambit of organic certification.
- NPOP has equivalence with organic food regulations of European Union and Switzerland and a recognition agreement on conformity assessment with USDA.
- TRACENET is an on-line application tool managed by APEDA under NPOP for management of entire organic certification system in the country.
- NPOP covers standards for crops and their products, live stocks and poultry products, aquaculture, apiculture etc. The exports from the country are as per the provisions in NPOP.

National Research Centres - 15

1.	<u>ICAR-National Research Centre for Banana, Trichi</u>
2.	<u>ICAR-National Research Centre for Grapes, Pune</u>
3.	<u>ICAR-National Research Centre for Litchi, Muzaffarpur</u>
4.	<u>ICAR-National Research Centre for Pomegranate, Solapur</u>
5.	<u>ICAR-National Research Centre on Camel, Bikaner</u>
6.	<u>ICAR-National Research Centre on Equines, Hisar</u>
7.	<u>ICAR-National Research Centre on Meat, Hyderabad</u>
8..	<u>ICAR-National Research Centre on Mithun, Medziphema, Nagaland</u>
9.	<u>ICAR-National Research Centre on Orchids, Pakyong, Sikkim</u>
10.	<u>ICAR-National Research Centre on Pig, Guwahati</u>

11.	<u>ICAR-National Research Centre on Plant Biotechnology, New Delhi</u>
12.	<u>ICAR-National Research Centre on Seed Spices, Ajmer</u>
13.	<u>ICAR-National Research Centre on Yak, West Kemang</u>
14.	<u>ICAR-National Centre for Integrated Pest Management, New Delhi</u>
15.	National Research Centre on Integrated Farming (ICAR-NRCIF), Motihari

Central Poultry Performance Testing Centre, Gurgaon (Haryana)

Central Sheep Breeding Farm, Hisar (Haryana)

Central Institute of Coastal Engineering For Fishery, Bengaluru

Central Institute of Fisheries, Nautical and Engineering Training, Cochin

National Institute of Fisheries, Post Harvest Technology and Training, Cochin

Fishery Survey of India, Mumbai.

National Institute of Animal Health, Baghpat (Uttar Pradesh).

National Institute of Agricultural Extension Management, Hyderabad

International Rice Research Institute (IRRI) South Asia Regional Centre established at Varanasi, U.P.

International institution namely International Center for Research in Semi Arid and Tropics (ICRISAT) in Hyderabad

National Institute of Plant Health Management Hyderabad

Central Insecticides Laboratory Faridabad

Central Institute of Horticulture (CIH), **Medziphema** Nagaland

Directorate of Economics & Statistics, an attached office of DAC&FW, is guided by the vision of enriching economic and statistical data and analytical inputs required by the Department of Agriculture, Cooperation & Farmers Welfare for providing better agricultural economic development policy. Its mission is to provide important statistics on area, production, cost and yield of principal crops, minimum support prices, to implement schemes related to improvement of agricultural statistics, and carrying out agro-economic research. The main objectives of the Directorate are generation and dissemination of agricultural statistics and research and analysis.

Agricultural Statistics (AS) Division of **Directorate of Economics & Statistics (DES)**, releases four Advance Estimates (AEs) followed by Final Estimates of production of major agricultural crops of the country. Each of these five estimates is available State-wise and at the national level for the 28 identified crops including pulses.

The First Advance Estimates are released in September. These cover only kharif crops, when kharif sowing is generally over.

The Second Advance Estimates are normally released in February, the following year when rabi sowing is also over. The second advance estimates cover kharif as well as rabi crops.

The Third Advance Estimates incorporate revised data on area coverage for rabi crops and better yield estimates of kharif crops. These are released in April-May.

The Fourth Advance Estimates are released in July-August.

Cost Study (CS) Division: Cost Study Division is mainly responsible for implementation and monitoring of a Comprehensive Scheme for Studying the Cost of Cultivation of Principal Crops in India and other issues relating to this scheme including answering parliament questions, generating cost of cultivation and production estimates and providing to CACP for the purpose of arriving at the recommendations of the Minimum Support Prices (MSP) etc.

Food Economics and Commercial Crops Divisions:

Food Economics Division of DES examines the Kharif and Rabi Price Policy Report of Commission for Agricultural Costs & Prices (CACP).

Commission for Agricultural Costs & Prices (CACP): The Commission for Agricultural Costs & Prices (CACP) is an attached office of the Ministry of Agriculture and Farmers Welfare, Government of India. It came into existence in January 1965 **based on recommendations of Jha Committee**. Currently, the Commission comprises a Chairman, Member Secretary, one Member (Official) and two Members (Non-Official). The non-official members are representatives of the farming community and usually have an active association with the farming community.

CACP recommends MSP for **22 crops and Fair and Remunerative prices of sugarcane** before the sowing period each year.

Also MSP derived for Toria based on MSP for Rapeseeds & Mustard and for De-husked Coconut on the Basis of MSP of Copra.

While recommending price policy of various commodities under its mandate, the Commission keeps in mind the various Terms of Reference (ToR) given to CACP in 2009. Accordingly, it analyzes:

- 1) demand and supply;
- 2) cost of production;
- 3) price trends in the market, both domestic and international;
- 4) inter-crop price parity;
- 5) terms of trade between agriculture and non-agriculture;
- 6) a minimum of 50 percent as the margin over cost of production; and
- 7) likely implications of MSP on consumers of that product.

It may be noted that cost of production is an important factor that goes as an input in determination of MSP, but it is certainly not the only factor that determines MSP.

MSP Recommendation:

CACP submits its recommendations to the government in the form of Price Policy Reports every year, separately for five groups of commodities namely **Kharif crops, Rabi crops, Sugarcane, Raw Jute and Copra.**

Before preparing aforesaid five pricing policy reports, the Commission draws a comprehensive questionnaire, and sends it to **all the state governments and concerned National organizations and Ministries to seek their views.**

Subsequently, separate meetings are also held with farmers from different states, state governments, National organizations like **FCI, NAFED, Cotton Corporation of India (CCI), Jute Corporation of India (JCI), trader's organizations, processing organizations, and key central Ministries.**

The Commission also makes visits to states for on-the-spot assessment of the various constraints that farmers face in marketing their produce, or even raising the productivity levels of their crops.

Based on all these inputs, the Commission then finalizes its recommendations/reports, which are then submitted to the government.

The government, in turn, circulates the CACP reports to state governments and concerned central Ministries for their comments. After receiving the feed-back from them, the Cabinet Committee on Economic Affairs (CCEA) of the Union government takes a final decision on the level of MSPs and other recommendations made by CACP.

Once this decision is taken, CACP puts all its reports on the web site for various stakeholders to see the rationale behind CACP's price and non-price recommendations.

Various committee related to MSP:

Y.K Alagh Committee (2005)

Hanumantha Rao Committee (1990)

S.R.Sen Committee Report (1979)

Jha Committee (1965)

Commercial Crops Division is responsible for examining the Price Policy Reports for Raw Jute and Copra submitted by CACP. The division also fixes the MSP of Toria by taking the average of price differentials between rapeseed/mustard and toria /lahi black in important markets for the previous five years.

Special Data Dissemination Standards (SDDS) Division: The Division is involved in collection and compilation of data relating to nine-fold classification of land, irrigated area (source wise and crop-wise) and total area under crops for States and UTs. The compiled data is brought out in the Directorate's publications of 'Land Use Statistics' and 'Agricultural Statistics at a Glance'.

Prices & Markets Division: Prices & Market Division provides necessary market intelligence to the Government and is involved in collection, compilation and dissemination of data/ information on prices - wholesale, retail, farm harvest and international prices of important agricultural commodities. Wholesale prices of 170

selected agricultural commodities from approx. 700 market centers spread across the country.

Soil and Land Use Survey of India (SLUSI), an apex organization in the country established in 1958 deals with soil survey and land resource mapping in the country. It is a subordinate field organization under Department of Agriculture, Cooperation and Farmers Welfare, Ministry of Agriculture and Farmers Welfare.

National Rainfed Area Authority (NRAA) was established as an attached office of DAC&FW. Dr. Ashok Dalwai (IAS) is leading the organization as its Chief Executive Officer (CEO). NRAA is an 'Advisory Body' for policy and programme formulation and monitoring of programmes/activities relating to integrated development of degraded/rainfed areas.

Agmark Grading Facilities:

Directorate of Marketing & Inspection (DMI) implements Agricultural Produce (Grading and Marking) Act, 1937 (as amended up to 1986) and other provisions as mandated by Food Safety Standards Regulations 2011 under Food Safety and Standards Act 2006. The agricultural produce are certified under AGMARK if they conform to grade standards under **Agricultural Produce (Grading Marking) Act, 1937**, General Grading and Marking Rules, 1988 (as amended up to 2009) and specific commodity Rules notified by DMI, Department of Agriculture, Cooperation and Farmers Welfare, Ministry of Agriculture & Farmers Welfare.

These standards differentiate between quality and 2-3 grades are prescribed for each commodity. Agmark standards for 224 agricultural commodities have been notified so far under the Provisions of the **APGM Act 1937** with an aim to provide premium quality products to the consumers, graded raw materials to processor/ manufacturers/ packers and remunerative prices to the farmers. The commodities are from group of Vegetables, cereals, pulses, oilseeds, vegetable oils, ghee, spices, honey, creamery butter, wheat, atta, besan, etc.

Area under organic certification (registered under NPOP) - **3.42 million Hectare.**

Directorate of Arecanut and Spices Development, Calicut: The Directorate of Arecanut and Spices Development (DASD) is a subordinate office under Ministry of Agriculture, Government of India, entrusted with the responsibility of development of Spices, Arecanut, Betel vine and Aromatic plants at National level. The Directorate monitors the development programmes implemented by the states in spices under MIDH.

Directorate of Cashewnut and Cocoa Development, Kochi: The Directorate of Cashewnut and Cocoa Development functioning at Kochi is the subordinate office of the Department of Agriculture, Cooperation & Farmers Welfare of the Union Ministry of Agriculture & Farmers Welfare.

The Directorate formulate and execute the development programmes of cashew and cocoa in the country and monitor the implementation of development programmes formulated and executed by the State Governments under Mission for Integrated Development of Horticulture.

The DCCD will also act as a nodal agency for accreditation of existing cashew/ cocoa nurseries and issue recognition. The formation of this Directorate was a breakthrough in the holistic development of these two versatile horticulture crops.

World Food Programme (WFP): Set up in 1963, WFP seeks to provide emergency feeding in places facing acute food insecurity due to natural calamities and man-made causes. Present share of Government of India for a biennium is US \$ 1.92 Million towards WFP Pledge Contribution which is used by them to support WFP India Country Programme.

International Fund For Agriculture Development (IFAD): India is a founder member of IFAD and a key contributor among the member Countries. IFAD funds are derived from member contributions (made in replenishment cycles), investment incomes and special funds. India has so far pledged US\$ 211 million and contributed US\$ 171 million to IFAD's Resources.

Revolutions related to agriculture:

❖ Black Revolution	Petroleum Production/crude oil production
❖ Brown Revolution	Leather/Cocoa production
❖ Golden Fibre Revolution	Jute Production
❖ Golden Revolution	Fruits/Overall Horticulture development/Honey Production
❖ Grey Revolution	Fertilizer
❖ Brown Revolution	Leather/Cocoa production
❖ Yellow Revolution	Oil Seeds
❖ Round Revolution	Potato
❖ Silver Fiber Revolution	Cotton
❖ Silver Revolution	Egg/Poultry
❖ Evergreen/Rainbow revolution	Overall development of Agriculture
❖ Sweet Revolution	Honey Production

❖ Green Revolution	Over all crop production
❖ White Revolution	Milk production

Five Year Plans (Concept taken from Russia)

FYP	Priority Area
**I FYP (1951-56) Harrod-Domar model	Famous called as “ Harrod Domar ” model, priority to agriculture and allied sectors
II FYP (1956-61) Mahalanobis Model	Famously called as Mahalanobis Model , Priority to Industry or manufacturing sector
**III FYP (1961-66)	Agriculture registered Negative Growth rate, priority to achieve Self sufficiency in food grain production and to stop import of food grains from USA
IV FYP (1969-74)	Growth and Justice
***V FYP (1974-79)	Removal of poverty and self reliance
VI FYP (1980-85) Rolling plan	Family planning
VII FYP (1985-90)	Food Production, Employment, Productivity
**VIII FYP (1992-97) Indicative planning	Famously called as “Indicative planning”, Employment Generation, Control of Population and Economic reforms
IX FYP (1997-2002)	Growth Rate of 7 percent
XFYP (2002-2007)	Self employment and resources and development
XI FYP (2007-2012)	Improvement of Health, Education and Sanitation.
XII FYP (2012-17)	Faster, Sustainable & more inclusive growth".

Some Important National Institutes:

INSTITUTE	Location
National Research Centre on Integrated Farming System (ICAR-NRCIF)	Motihar, Bihar
IRRI South Asia Regional Centre	Varanasi, U.P.
*Indian Institute of Farming Systems Research (IIFSR)	Meerut, UP
**IIRS – Indian Institute of Remote Sensing,	Dehradun, Uttarakhand
**CIRCOT – Central Institute for Research on Cotton Technology,	Mumbai, Maharashtra
CIRG – Central Institute for Research on Goats,	Makhdoom, Farah, Uttar Pradesh
*Indian Institute of Agricultural Biotechnology (IIAB)	Ranchi, Jharkhand
*National Institute of Biotic Stresses Management (NIBSM)-	Raipur, Chhattisgarh
*National Institute of Abiotic Stress Management (NIASM) -	Malegaon, Maharastra
CISTH – Central Institute of Sub-Tropical Horticulture	Lucknow, Uttar Pradesh
CITH – Central Institute of Temperate Horticulture	Srinagar, Jammu & Kashmir
central Institute of Arid Horticulture	Bikaner, rajasthan
Central Seed Testing Referral Laboratory	Varanasi
**CIWA - Central Institute for Women in Agriculture	Bhubaneshwar, Odisha

CIP (International Potato Center: Lima, Peru
International Horticulture Innovation And Training Center: Jaipur, Rajasthan
ILRI (International Livestock Research Institute): Nairobi, Kenya
IRRI - (International Rice Research Institute) : Los Banos, Philippines
Food and Agriculture organization (FAO): Rome, Italy

NATIONAL BUREAUS- 06

NBAGR – National Bureau of Animal Genetic Resources	Karnal, Haryana
NBAII – National Bureau of Agriculturally Important Insects	Hebbal, Bengaluru, Karnataka
NBAIMO – National Bureau of Agriculturally Important Micro-Organisms	Mau Nath Bhanjan, Uttar Pradesh
NBFGR – National Bureau of Fish Genetic Resources	Lucknow, Uttar Pradesh
NBPGR – National Bureau of Plant Genetic Resources	New Delhi
NBSS & LUP – National Bureau of Soil Survey and Land-Use Planning	Nagpur, Maharashtra

Deemed Universities- 04

IARI - Indian Agricultural Research Institute	New Delhi
NDRI - National Dairy Research Institute	Karnal, Haryana
IVRI - Indian Veterinary Research Institute	Izatnagar, Uttar Pradesh
CIFE - Central Institute on Fisheries Education	Mumbai, Maharashtra

Food Corporation of India: The Food Corporation of India was setup under the Food Corporation's Act 1964 , in order to fulfill following objectives of the Food Policy:

- Effective price support operations for safeguarding the interests of the farmers.
- Distribution of foodgrains throughout the country for public distribution system.
- Maintaining satisfactory level of operational and buffer stocks of foodgrains to ensure National Food Security

Since its inception, FCI has played a significant role in India's success in transforming the crisis management oriented food security into a stable security system.

It was set up in 1965 and headquarter New Delhi.

The Food and Agriculture Organization of the United Nations (FAO)https://en.wikipedia.org/wiki/Food_and_Agriculture_Organization - [cite note-1](#) is a specialized agency of the United Nations that leads international efforts to defeat hunger and improve nutrition and food security. Its Latin motto, fiat panis,

translates to "let there be bread". Founded in October 1945, the FAO is the oldest existing agency of the U.N

The FAO is headquartered in Rome, Italy.

The Rubber Board is a statutory body managed by the Ministry of Commerce and Industry, under the Rubber Act 1947, for the overall development of the rubber industry in the country.

Head Office is in Kottayam, Kerala

The Coffee Board of India is an organisation managed by the Ministry of Commerce and Industry of the government of India to promote coffee production in India. Head Office is in Bangalore.

The activities of the Board are broadly aimed at (i) enhancement of production, productivity & quality; (ii) export promotion for achieving higher value returns for Indian Coffee and (iii) supporting development of Domestic market.

Spices Board: Spices Board (Ministry of Commerce and Industry, Government of India) is the flagship organization for the development and worldwide promotion of Indian spices. The Board is an international link between the Indian exporters and the importers abroad. The Board has been spearheading activities for excellence of Indian spices, involving every segment of the industry. The Board has made quality and hygiene the corner stones for its development and promotional strategies. The board is headquartered in Kochi.

The Tea Board of India is a state agency of the Government of India established to promote the cultivation, processing, and domestic trade as well as export of tea from India. It was established by the enactment of the Tea Act in 1953 with its headquarters in Kolkata (formerly Calcutta)

TOBACCO BOARD: Recognizing the need to regulate production, promote overseas marketing and control recurring instances of imbalances in supply and demand, which lead to market problems, the Government of India under the Tobacco Board Act of 1975, established the Tobacco Board, in place of the Tobacco Export Promotion Council. The Board came into existence from 1-1-1976 and opened its **headquarters at Guntur in Andhra Pradesh, India.**

The Tobacco Board Act, 1975 aims at planned development of Tobacco Industry in the country.

Miscellaneous Topics for Agriculture

Branches of Agriculture

Agriculture is the branch of science encompassing the applied aspects of basic sciences. The applied aspects of agricultural science consist of study of field crops and their management (Aviculture) including soil management. There are a large number of fields under agriculture, as under.

Plant Breeding/Genetics is the art and science of changing the traits of plants in order to produce desired characteristics. Plant breeding can be accomplished through many different techniques ranging from simply selecting plants with desirable characteristics for propagation, to more complex molecular techniques (see cultigen and cultivar). Plant breeding has been practiced for thousands of years, since near the beginning of human civilization.

Agronomy is the science and technology of producing and using plants for food, fuel, fibre, and land reclamation. Agronomy encompasses work in the areas of plant genetics, plant physiology, meteorology, and soil science. Agronomy is the application of a combination of sciences like biology, chemistry, economics, ecology, earth science, and genetics. Agronomists today are involved with many issues including producing food, creating healthier food, managing environmental impact of agriculture, and extracting energy from plants.

Horticulture is the other important section of agriculture which comes from the Latin hortus “garden” and cultūra “cultivation”, from cultus, the perfect passive participle of the verb colō “I cultivate”. Hortus is cognate with the native English word yard (in the meaning of land associated with a building) and also the borrowed word garden. Cultivation and intensive focus to crops like mango, banana, sapota, guava and vegetables like tomato, potato, onion comes under this section of agriculture.’

Divisions of Horticulture:

Horticulture crops include fruits, vegetables, flowers, plantation crops, Spices, condiments, Medicinal and Aromatic crops etc.

Pomology: It is derived from two words i.e. ‘Pomum’ meaning fruit and ‘Logos’ meaning discourse or study. So, Pomology is study or cultivation of fruit crops. For example, mango, sapota, guava, grape, banana etc.

Olericulture: It is derived from two words ie. ‘Oleris’ meaning Potherb and ‘Cultra’ meaning cultivation. So, Olericulture literally means potherb cultivation. In the present days it is broadly used to indicate the cultivation of vegetables. For example, brinjal, okra, tomato, pumpkin etc.

Floriculture: It is derived from two words i.e. ‘Florus’ meaning flower and ‘Cultra’ meaning cultivation. So floriculture means study of flower crops. In this there are again two subdivisions.

(1) Commercial Floriculture (2) Ornamental Floriculture.

Commercial Floriculture: Deals with the cultivation of flower crops grown on commercial scale for profit (Income). For example, rose, jasmine, carnation, aster, and marigold etc.

Ornamental Floriculture: It deals with the raising of flower crops for ornamental, pleasure and fashion purposes. For example, dahlia, zinnia, cosmos, hibiscus, balsam, nerium, poinsettia, hollyhock, gerbera, and gaillardia etc.

Arboriculture: This branch deals with the raising of perennial trees meant for shade, avenue or ornamental purposes. For example, polyalthia, spathodea, cassia, gulmohar etc.

Plantation Crops: Are those crops, which are cultivated in an extensive scale in large contiguous areas, owned and managed by an individual or a company and whose produce is utilized only after processing. For example, coffee, tea, rubber, coconut, cocoa etc.

Spices and Condiments: This branch deals with the cultivation of crops whose produce is used mainly for seasoning and flavouring dishes.

Spices are those plants, the products of which are made use of as food adjuncts to add aroma and flavour. For example, pepper, cardamom, clove, cinnamon, All spice etc.

Condiments are those plants, the products of which are made use of as food adjuncts to add taste only. For example, turmeric, ginger, red chillies, onion, garlic etc. Both spices and condiments contain essential oils, which provide aroma, flavour and taste and they are of little nutritive value.

Medicinal and Aromatic Plants: It deals with the cultivation of medicinal plants, which provide drugs and aromatic crops which yields aromatic (essential) oils.

Medicinal Plants: are those plants, which are rich in secondary metabolites and are potential sources of drugs. The secondary metabolites include alkaloids, glycosides, coumarins, flavonoides and steroids etc. For example, periwinkle, opium, methi, cinchona, dioscorea yam, belladonna, senna, sarpagandha, aswagandha, tulsi etc.

Aromatic Plants: are those plants, which possesses essential oils in them. The essential oils are the odoriferous steam volatile constituents of aromatic plants. For example, lemon grass, citronella, palmarosa, vetiver, geranium, davanam, lavender etc.

Post Harvest Technology: It deals with the processing and preservation of produce of horticulture crops.

Landscape Gardening: It deals with the planning and execution of ornamental gardens, parks, landscape gardens etc.

Nursery and Seed Production: It deals with the production of seeds and planting material of horticulture crops on commercial basis. Plants grown for aesthetic value are also included in horticulture. Though crops like potato, cowpea and several condiments are grown as field crops they are included under horticultural crops when they are grown as vegetables in small areas.

Horticulture also includes the fields like mushroom cultivations which is concerned with cultivation of mushroom in a controlled and intended manner to use it for the purpose of food especially button mushroom i.e. *Agaricus bisporus*, which is common in cultivation in India.

Agroforestry is the science dealing with land use management system of growing shrubs, trees around or among crops or pastureland. In simple words it combines agricultural and

forestry technologies to create more diverse, productive, profitable, healthy, system for farming. Most common plants used in the agroforestry are subabul, pongamia, jatropa etc.

Forestry It deals with the large scale cultivation of perennial trees for supplying wood, timber, rubber, etc. and also raw materials for industries.

Soil Science is the study of soil as a natural resource on the surface of the Earth including soil formation, classification and mapping; physical, chemical, biological, and fertility properties of soils; and these properties in relation to the use and management of soils. Sometimes terms which refer to branches of soil science, such as *pedology* (formation, chemistry, morphology and classification of soil) and *edaphology* (influence of soil on organisms, especially plants), are used as if synonymous with soil science.,

Agricultural Microbiology is a branch of microbiology dealing with plant-associated microbes and plant and animal diseases. It also deals with the microbiology of soil fertility, such as microbial degradation of organic matter and soil nutrient transformations.

Agricultural Biotechnology also known as agritech, is an area of agricultural science involving the use of scientific tools and techniques, including genetic engineering, molecular markers, molecular diagnostics, vaccines, and tissue culture, to modify living organisms: plants, animals, and microorganisms.

Agricultural Engineering is the engineering discipline that applies engineering science and technology to agricultural production and processing. Agricultural engineering combines the disciplines of mechanical, civil, electrical and chemical engineering principles with a knowledge of agricultural principles.

Agricultural Extension is a general term meaning the application of scientific research and new knowledge to agricultural practices through farmer education. The field of 'extension' now encompasses a wider range of communication and learning activities organized for rural people by educators from different disciplines, including agriculture, agricultural marketing, health, and business studies. Extension practitioners can be found throughout the world, usually working for government agencies. They are represented by several professional organizations, networks and extension journals.

Agricultural Economics: It is an applied field of economics concerned with the application of economic theory in optimizing the production and distribution of food and fibre — a discipline known as agronomics. Agronomics was a branch of economics that specifically dealt with land usage. It focused on maximizing the crop yield while maintaining a good soil ecosystem.

Dairying: The human need for milk is taken care by the field of dairying which refers to longterm availability and production of milk, which is processed further for eventual meeting the need of milk of the society. Some of the most common named cow breeds for milk in India are Gir, Red Sindhi, Sahiwal, Tharparkar, Rath etc...

Poultry is the branch of agriculture which deals with the rearing birds kept for meeting the food demand getting the eggs they produce, their meat, their feathers. In India some of the reared poultry breeds are Aseel, Karaknath, Naked Neck, Sonali, Devendra, Priya, Tennis, Punjab, Brown, Ghagus, Lolab, Kashmir, *Leghorn* etc...

Piggery is the field which concerns the farm rearing and breeding pigs for the livestock purpose, Pigs convert inedible feeds, forages, certain grain byproducts obtained from mills, meat by products, damaged feeds and garbage into valuable nutritious meat. Most of these

feeds are either not edible or not very palatable to human beings. Some of the pig breeds reared in India are Large white Yorkshire, Land race, Middle white Yorkshire, Hampshire, Duroc etc.

Apiculture or Beekeeping denotes the rearing and maintenance of honey bee colonies, commonly in natural and artificial hives for the purpose of getting honey and bee products. Bee species like *Apis florea*, *Apis dorsata*, *Apis mellifera* are commonly known for rearing, in all these *Apis mellifera* is most commonly reared species.

Sericulture is a branch of agriculture dealing with the silk farming, that is rearing of silkworms for the production of silk, especially the species *Bombyx mori* L., which is commercially used in Indian regions like Assam, Meghalaya, that are of prime importance in the field of sericulture.

Agriculture Entomology is the branch dealing with study of agriculturally important insects and pest, While, *Soil Microbiology* deals with soil property and health, water management plans such as irrigation and water use systems are also equally important for the holistic development of agriculture.

Nematology is the scientific discipline concerned with the study of nematodes, or roundworms. Although nematological investigation dates back to the days of Aristotle or even earlier, nematology as an independent discipline has its recognizable beginnings in the mid to late 19th century

Plant Pathology (also *Phytopathology*) is the scientific study of diseases in plants caused by pathogens (infectious organisms) and environmental conditions (physiological factors). Organisms that cause infectious disease include fungi, oomycetes, bacteria, viruses, viroids, virus-like organisms, phytoplasmas, protozoa, nematodes and parasitic plants.

Animal Husbandry: Maintenance of various types of livestock for direct energy (work energy), milk and meat. Husbandry is for direct and indirect energy.

Fishery Science : It is for marine fish and inland fishes including shrimps and prawns.

Home Science : Application and utilization of agricultural produces in a better manner.

Organic agriculture Organic agriculture is a production system that sustains the health of soils, ecosystems and people. It relies on ecological processes, biodiversity and cycles adapted to local conditions, rather than the use of inputs with adverse effects. Organic agriculture combines tradition, innovation and science to benefit the shared environment and promote fair relationships and a good quality of life for all involved.

Principles of Organic Agriculture

Principle of health: Organic agriculture sustains and enhances the health of soil, plant, animal, human and planet as one and indivisible.

Principle of ecology: Organic agriculture is based on living ecological systems and cycles; it works with, emulates and helps sustain them.

Principle of fairness: Organic agriculture builds on relationships that ensure fairness with regard to the common environment and life opportunities.

Principle of care: Organic agriculture is managed in a precautionary and responsible manner to protect the health and well-being of current and future generations and the environment.

Components of Organic Farming:

Crop Rotation : It is a systematic arrangement of growing different crops in a more or less regular sequence on the same land covering a period of two years or more. Through crop rotations, soil fertility, weeds, insects and diseases are managed without any chemical inputs.

Crop Residue : In India there is a great potential for utilization of crop residues/ straw of some of the major cereals and pulses. About 50% of the crop residues are utilized as animal feed, the rest could be very well utilized for recycling of nutrients with the help of efficient microbial inoculants.

Organic Manure : The organic manure is derived from biological sources like plant, animal and human residues. Organic manure act in many ways in augmenting crop growth and soil productivity:

a) Bulky Organic Manure: It generally contains less amounts of plant nutrients as compared to concentrated organic manure. It includes farm yard manure (FYM), compost and green manure.

- **FYM:** It refers to the well-decomposed mixture of dung, urine, farm litter and left over or used up materials from roughages or fodder fed to the cattle.
- **Compost:** Large quantities of waste material are available as vegetable refuse, farm litter, such as weeds, stubble, straw, sugarcane trash, sewage sludge and animal waste in houses and in areas like human and industrial refuse; therefore, excreta can be converted into useful compost manure by conserving and subjecting these to a controlled process of anaerobic decomposition. Compost is used in the same way as FYM and is good for application to all soils and all crops.
- **Green Manuring:** It is a practice of ploughing or turning into the soil undecomposed green plant tissues for the purpose of improving physical structure as well as fertility of the soil. The most commonly used green manuring crops are: Sunhemp (*Crotalaria juncea*), Dhaincha (*Sesbania aculeata*), Clusterbean (*Cyamopsis tetragonoloba*), Senji (*Melilotus parviflora*), Cowpea (*Vigna catjang*, *Vigna sinensis*), Berseem (*Trifolium alexandrium*).

b) Concentrated Organic Manure: Concentrated organic manures are those materials that are organic in nature and contain higher percentage of essential plant nutrients such as nitrogen, phosphorous and potash, as compared to bulky organic manures. These concentrated manures are made from raw materials of animal or plant origin. The concentrated organic manures commonly used are oilcakes, blood meal, fishmeal, meat meal and horn and hoof meal.

Waste as Organic Manures:

- **Industrial waste:** Among the industrial by-products, spent wash from dertilisers and molasses and pressmud from sugar industry have good manurial value. Addition of

decomposed pressmud improves the soil fertility and enhances the activity of microbes. Coir waste is the by-product from coir industry and can be used as manure after proper decomposition.

- **Municipal and Sewage waste:** It also forms an important component of organic waste. Sewage sludge particularly from industrialized cities is contaminated with heavy metals and these pose hazards to plants, animals and human beings. Thus, toxic waste should be separated at the source to minimize the concentration of such elements in the sludge.

Biofertilizers Bio-fertilizer is microorganism's culture capable of fixing atmospheric nitrogen when suitable crops are inoculated with them. The main inputs are microorganisms, which are capable of mobilizing nutritive elements from non-usable form to usable form through biological process. These are less expensive, eco-friendly and sustainable. The Biofertilizers containing biological nitrogen fixing organism are more important in agriculture. Bio-fertilizers may be of two types:

- **Symbiotic N-fixation** where Rhizobium culture of various strains is used. Rhizobium culture multiplies in the roots of legume crops and fixes nitrogen symbiotically.
- **Asymbiotic N-fixation** with help of Azotobacter, Azospirillum, Blue Green Algae, Azolla and Mycorrhizae. They grow on decomposing soil organic matter and produce nitrogen compounds.

Different Types of Biofertilizers: The following microorganisms are used as biofertilizers:

- **Rhizobium:** They form root nodules in leguminous plants and fix the atmospheric nitrogen into an organic form. Rhizobium also has no negative effect on soil quality and improves the quality, nutrient content, and growth of the plant. This is the most common biofertilizer as stated earlier. Rhizobium lives in the root hairs of the legumes by forming nodules.
- **Azotobacter:** These are free-living nitrogen fixers found in all types of upland crops. These not only fix nitrogen but also provide certain antibiotics and growth substances to the plant. Azotobacter is a heterotrophic free living nitrogen fixing bacteria present in alkaline and neutral soils. Azotobacter chroococcum is the most commonly occurring species in arable soils of India. Apart from its ability to fix' atmospheric nitrogen in soils, it can also synthesize growth promoting substances viz., auxins, and gibberellins and also to some extent the vitamins.
- **Acetobacter** Acetobacter diazotrophicus is a newly discovered nitrogen fixing bacteria associated with sugarcane crop. This bacterium belongs to the alpha group of proteobacteria. It was isolated from leaf, root, bud and stem samples of sugarcane.
- **Azospirillum:** Unlike Azotobacter, these can be used in wetland areas. They are found inside the roots of the plant (non-free-living) where they fix the atmospheric nitrogen. This is a free living or non -symbiotic bacteria (does not form nodules but makes association by living in the rhizosphere). Azospirillum species establish an association with many plants particularly with C, plants such as maize, sorghum, sugarcane, etc. It is the most common organism and can form associative symbiosis on a large variety of plants.

The bacteria live on root surface, sometimes also penetrates into the root tissues but do not produce any visible nodule or out growth on the root tissue. They fix nitrogen from 10 to 40 kg/ha.

- **Blue-green algae:** These are free-living nitrogen-fixing Cyanobacteria that are present only in wet and marshy lands. However, they do not survive in acidic soil. Bluegreen algae are also known by different nomenclature such as, cyanophytes, cyanobacteria and most recently cyanoprokaryotes. The Blue-Green Alga (*Anabaena azollae*) forms a symbiotic relationship with *Azolla* (aquatic fern) and fixes atmospheric nitrogen.
- **Azolla:** *Azolla* is a free-floating water fern that floats in water and fixes atmospheric nitrogen in association with nitrogen fixing blue green alga *Anabaena azollae*. *Azolla* fronds consist of sporophyte with a floating rhizome and small overlapping bi-lobed leaves and roots.
- **Frankia** *Frankia* is actinomycetes which also fixes atmospheric nitrogen. It forms a symbiotic association by forming root nodules in some non-leguminous trees such as *Casuarina* and *Alnus*. The wasteland soil fertility can be improved by growing *Casuarina*. *Frankia* makes *casuarina* tree suitable for agro-forestry system in nitrogen deficient soils

Phosphorous Fixing Fertilizers:

- **Mycorrhiza:** It is a symbiotic association between the fungi and the roots of a plant. The mycorrhizal fungi play an important role in binding the soil together and improves the activity of the microbes. The fungi draw water and nutrients from the soil thereby increasing the plant productivity. It also helps the plant to survive under various environmental stresses and help in nutrient transfer mainly of phosphorus, zinc and sulfur.
- **Phosphate Solubilizing Microorganisms (PSM):** Several soil bacteria and fungi, notably species of *Pseudomonas*, *Bacillus*, *Penicillium*, *Aspergillus* etc. secrete organic acids and lower the pH in their vicinity to bring about dissolution of bound phosphates in soil. Increased yields of wheat and potato were demonstrated due to inoculation of peat based cultures of *Bacillus polymyxa* and *Pseudomonas striata*.

Applications of Biofertilizers: Following are the important applications of biofertilizers:

Seedling root dip: This method is applicable to rice crops. The seedlings are planted in the bed of water for 8-10 hours.

Seed Treatment: The seeds are dipped in the mixture of nitrogen and phosphorus fertilizers. These seeds are then dried and sown as soon as possible.

Soil Treatment: The biofertilizers along with the compost fertilizers are mixed and kept for one night. This mixture is then spread on the soil where the seeds have to be sown.

Bio-pesticide Bio-pesticides are natural plant products that belong to the so-called secondary metabolites, which include thousands of alkaloids, terpenoids, phenolics and minor secondary chemicals and are effective against many insects, nematodes, fungi etc. *Neem*-(*Azadirachta indica*) is one of the botanical pesticides used widely. In fact, all parts of the *Neem* tree possess insecticidal property but seed kernel is most active. Some other

commonly used botanical insecticides are Nicotine, Pyrethrum, Rotenone, Subabilla, Ryanin, Quassia, Margosa, Acorus, etc.

Vermicompost As the name indicates, vermicompost- organic manure is produced by the activity of earthworms. Vermicomposting is a method of making compost with the use of earthworms that generally live in soil, eat biomass and excrete it in digested form. These are rich in macro and micronutrients, vitamins, growth hormones and immobilized microflora.

Weed Management In organic farming, weeds are managed by promoting crop competition and phytotoxic effects on weeds. Organic farmers integrate cultural, biological, mechanical, physical and chemical tactics to manage weeds without synthetic herbicides. Mechanical and physical weed control practices used on organic farms can be broadly grouped as:

- **Tillage** - Turning the soil between crops to incorporate crop residues and soil amendments; remove existing weed growth and prepare a seedbed for planting; turning soil after seeding to kill weeds, including cultivation of row crops
- **Mowing and cutting** - Removing top growth of weeds
- **Flame weeding and thermal weeding** - Using heat to kill weeds
- **Mulching** - Blocking weed emergence with organic materials, plastic films, or landscape fabric.

Types of Organic Fertilizers: Organic fertilizers come in the different forms. The most common organic fertilizer is animal manure. Fish emulsion, bone meal, cottonseed meal, blood meal and sludge are other forms of organic fertilizer.

Advantages: In the case of both kinds of fertilizers, using them in the correct amounts leads to more fertile soil. Some of the specific advantages of organic fertilizers, like sludge or manure, include improved soil texture, water retention and resistance to erosion. Inorganic fertilizers.

Genetically modified crops Organic agriculture does not allow any genetically modified (GM) organisms in farming system.

Benefits of Organic Farming:

- It helps in maintaining environment health by reducing the level of pollution.
- It reduces human and animal health hazards by reducing the level of undesirable and toxic residues in the product.
- It helps in keeping agricultural production at a higher level and makes it sustainable.
- It reduces the cost of agricultural production and also improves the soil health.
- It ensures optimum utilization of natural resources for short-term benefit and helps in conserving them for future generation.
- It not only saves energy for both animal and machine, but also reduces risk of crop failure. It improves the soil physical properties such as granulation, and good tilth, good aeration, easy root penetration and improves water-holding capacity.
- It improves the soil's chemical properties such as supply and retention of soil nutrients, and promotes favorable chemical reactions.

Natural farming is an ecological farming approach established by Masanobu Fukuoka (1913– 2008), a Japanese farmer and philosopher, introduced in his book 'The One-Straw Revolution' in 1975. It is also referred to as “the Fukuoka Method”, “the natural way of farming” or “donothing farming”. Natural farming is related to fertility farming, organic farming, sustainable agriculture, agroforestry, ecoagriculture and permaculture but should be distinguished from biodynamic agriculture. The most essential aspect of natural farming is to let nature play a dominant role to the maximum extent possible. Hence, no-till, farm biodiversity, integration and symbiotic farm components and protection of soil cover all have a place in this method of farming.

In India, natural farming is often referred to as “Rishi Kheti” which includes ancient vedic principles of farming based on the use of animal waste and herbs for controlling pests and cow products like buttermilk, milk, curd and its waste urine for preparing growth promoters. The Rishi or Vedic farming is regarded as non-violent farming without any usage of chemical fertilizer and pesticides. Principally, natural farming minimizes human labour and intervention and adopts, as closely as practical, nature’s production of foods.

Genetically modified crops Genetically modified (GM) or biotech crops are improved cultivars developed using biotechnological procedures and defined as plants whose genetic material has been artificially modified by addition or deletion of gene in laboratory using genetic engineering/ recombinant DNA technology. Power of genetic engineering/recombinant DNA technology has facilitated the transfer of gene (s) from taxonomically different organism such as plant, animal, bacterial and viral genes or even gene synthesized in laboratory. This kind of genetic modification or gene transfer is not possible in nature or through traditional crossbreeding methods.

The gene used for modifying plant genome for development of GM crops is called transgene, if the gene is taken from the organism which is not crossable to target plant species. The crystal protein (cry) genes isolated from bacterium–*Bacillus thuringiensis* and transferred into many plant species including cotton and maize is an excellent example of transgene.

Crop plants modified with such gene (s) are called transgenic crop or genetically modified or biotech crops. The other possibility of GM crops is the modification of plant genome in the laboratory using gene from same or from a crossable species. Such gene (s) is called cisgene and the plant genome modified using cisgene is referred as cisgenic crop.

In India, Bt-cotton namely MECH-12, MECH-162 and MECH-184 were released for cultivation in 2002. The Bt GM cotton hybrids were consisted of a gene cry1Ac isolated from bacterium – *Bacillus thuringiensis*. Other GM crops include gene for herbicide tolerance, disease resistance, male sterility and fertility, quality improvement transferred in many crops.

GM crops approved in India: So far, four Bt. cotton hybrids have been approved for commercial cultivation by GEAC, MOEF. Out of four hybrids approved so far, Bt. MECH-12, Bt. MECH-162 and Bt. MECH-184 are of M/s. Mahyco – Monsanto and RCH 2 Bt of M/s. Rasi Seeds Pvt. Ltd. These hybrids are being cultivated in six states, namely Gujarat, MP, Maharashtra, AP, Karnataka and Tamil Nadu. North India likely to be released during 2005 However, transgenic mustard, corn, brinjal, tomato are also under various stages of testing and trials.

Certification Agencies: There are many agencies designated as certification agencies in the country. Some are government owned and others are franchise of international

certification agencies. All the certifications whether they are done by a government or by a non-government organisation are accredited as per the NPOP and APEDA recommendations. The number of agencies keep on changing due to addition and deletion in the list as per the approval from APEDA. The updated list of certification agencies is available on the APEDA website (www.apeda.gov.in).

Participatory Guarantee System (PGS): Participatory Guarantee System (PGS) is a quality assurance initiative implemented by the Ministry of Agriculture and Farmers Welfare, Government of India through National Centre for Organic Farming (NCOF). In PGS, people in similar situations (in case of small producers) assess, inspect and verify the production practices of each other and take decision on organic certification. PGS has a number of basic elements which embrace a participatory approach, a shared vision, transparency and trust. Participation is an essential and dynamic part of PGS.

Organisations Involved in Organic Farming:

- The Agricultural and Processed Food Products Export Development Authority (APEDA)
- National Centre for Organic Farming (NCOF)

The Agricultural and Processed Food Products Export Development Authority (APEDA)

The main functions of APEDA are –

- Promoting the export-oriented production and development of organic agri-products.
- Fixing the standards and specifications for the scheduled products for the purpose of export.

APEDA provides licence to certification agencies operating in the country. It also monitors them as and when required. The agricultural products are allowed to be exported as organic only if they fulfil the norms and standards prescribed for them.

Head office: New Delhi

National Centre for Organic Farming (NCOF): This is a training institute under the Ministry of Agriculture and Farmers Welfare, Government of India. There are six regional centres at Bengaluru, Bhubaneswar, Hisar, Imphal, Jabalpur and Nagpur.

The main functions of NCOF are –

- Promotion of organic farming in the country through technical capacity building of all stakeholders including human resource development, transfer of technology, promotion and production of quality organic and biological inputs.
- Certification of organic products. NCOF is also involved in awareness generation, capacity building, training and demonstrations, providing financial support to organic inputs and market development for farmers.

Government Initiatives: The Government of India is promoting organic farming through various schemes and programmes:

National Mission for Sustainable Agriculture (NMSA)

Mission for Integrated Development of Horticulture (MIDH)

Paramparagat Krishi Vikas Yojana (PKVY)

Mission Organic Value Chain Development for North-eastern Region (MOVCD NER)

National Programme for Organic Production (NPOP)—This programme by the Government of India provides an institutional mechanism for implementation of the

National Standards for Organic Production: National Standards for Organic Production (NSOP)— This body sets out the standards to be followed in the cultivation, harvest, production, processing and trading of organic products.

The Standards for Organic Production are notified in National Programme for Organic Production (NPOP) by the Director General of Foreign Trade under the Foreign Trade (Development and Regulation) Act, 1992.

The NPOP has a system where in Certification Bodies are accredited with National Accreditation Body (NAB) which is chaired by the Additional Secretary in the Department of Commerce.

The Secretariat of the NAB is with APEDA. NPOP covers standards for crops and their products, live stocks and poultry products, aquaculture, apiculture etc. The exports from the country are as per the provisions in NPOP.

The Government of India declared Sikkim as the first organic state on 18 January 2016 and it is celebrated as the Sikkim Organic Day.

Precision Farming (PF): Precision Farming (PF) is an approach to farm management that uses information technology (IT) to ensure that the crops and soil receive exactly what they need for optimum health and productivity. The goal of PF is to ensure profitability, sustainability and protection of the environment. PF is also known as satellite agriculture, as-needed farming and site-specific crop management (SSCM).

It includes the use of latest technologies such as remote- sensing (RS), GPS, and geographical information system (GIS) with an objective to improve profitability and productivity.

Precision Farming relies upon specialized equipment, software and IT services. The approach includes accessing real-time data about the conditions of the crops, soil and ambient air, along with other relevant information such as hyper-local weather predictions, labor costs and equipment availability. Predictive analytics software uses the data to provide farmers with guidance about crop rotation, optimal planting times, harvesting times and soil management.

Sensors in fields measure the moisture content and temperature of the soil and surrounding air. Satellites and robotic drones provide farmers with real-time images of individual plants. Information from those images can be processed and integrated with sensor and other data to yield guidance for immediate and future decisions, such as precisely what fields to water and when or where to plant a particular crop.

Protected cultivation can be defined as a cropping technique wherein the micro climate surrounding the plant body is controlled partially or fully to maintain optimum conditions for better growth and development of crops. Under protected cultivation, crops are grown inside a framed structure that will protect the crop from environmental extremes, excess rainfall, wind, insects-pests on one hand and on the other it create environmental conditions that is more conducive for plant growth and development.

Seed Seeds can be defined as a dormant embryo (microseedling), which develops into a plant when subjected to required environmental conditions

Parts of seed A seed comprises the following parts

- (a) Seed coat
- (b) Embryo: Cotyledons or endosperm
- (c) Plumule bud

(d) Radicle

Characteristics/Attributes of Quality Seed

1. Genetic purity: Genetic purity refers to the percentage of contamination by seeds or genetic material of other varieties or species. In general, the genetic purity of the seed planted must equal or exceed the final product purity standard required, as purity generally decreases with each subsequent generation of propagation. For example Breeder seed 100%, Foundation seed 99%, certified seed 98%.

2. Physical purity: Physical purity of seed is the proportion of pure seed component in the seed lot as well as the proportion of other crop seed, weed seed and inert matter. Physical purity for most of the crops should be 98% and seed lot should be free from other crop seeds (number/kg), designated inseparable crop seeds, weed seed and other field impurities

3. Germination percentage: The quality seed should have germination percentage according to the standard of Indian Minimum Seed Certification Standard (IMSCS) so that farmers can get optimum plant stand in the field.

4. Vigour: Seed vigour is the sum total of those properties of the seed, which determine the level of activity and performance of the seed or seed lot during germination and seedling emergence. In general, it is the potential of seed in terms of better germination, growth and finally good yield.

5. Viability: The viability of the seed is a measure of seed alive and could develop into plants, which will reproduce themselves, given the appropriate conditions. It is measured through tetrazolium chloride test.

6. Moisture content: The moisture content is the amount of water in the seed and is usually expressed as a percentage. A small change in seed moisture content has a large effect on the storage life of the seeds. Therefore, it is important to have optimum moisture content in good quality seed. For example, cereals: 10-12 %, pulses: 7-9% and oilseeds: 6-7%, vegetables: 5-6%.

Pollination: It is the process by which pollen is transferred from the anther (male part) to the stigma (female part) of the plant, thereby enabling fertilization and seed formation inside the ovary.

Types of pollination Based on transfer of pollen from anthers to stigma, pollination is classified into two groups

1. Self-pollination/ autogamy: When pollen grains are transferred from an anther to the stigma of the same flower, the process is called self-pollination.

2. Cross-pollination/allogamy: If pollen grains are transferred to the stigma of another flower, then process is known as cross pollination.

Based on pollination behaviour crops are classified under following three categories.

I. Self-pollinated crops eg. wheat, rice, most of the pulse crops

II. Offen cross-pollinated crops eg. arhar, castor

III. Cross-pollinated crops eg. maize, bajra

Seed Development After fertilization, development of fertilized ovule into a mature seed involves several different stages. Seed formation begins within the minute embryo sac with

certain expectations, which is about the same in shape, size, and arrangement. In spite of initial similarities, the seed develops according to the genetic specification for each species, which are coded in the nucleus (chromosomes) of each cell.

Variety: Variety is the group of plants, which have similar morphological characters and belongs to same genus and species.

Hybrid: Hybrid is the product of two dissimilar parents or lines of dissimilar genotype in order to combine desirable characters from both the parents.

Inbred: It is plant type in cross pollinated crops, which is produced through continuous self-pollination process, which is further used as parental line for hybrid formation.

Stages of Seed Multiplication After the development and release of plant variety, the initial amount of pure seed, which is limited, should be multiplied through scientifically methods under different stages and these stages are called classes of seed multiplications. The following four different classes are used for multiplication of newly released variety.
a. Nucleus seed b. Breeders seed c. Foundation seed d. Certified seed

Nucleus Seed :- It is the initial amount of pure seed of improved variety or parental lines of a hybrid produced under supervision of the plant breeder who has evolved that variety or hybrid. The nucleus seed is genetically cent percent pure and does not contain other physical impurities. The nucleus seed is produced strictly under isolation so as to avoid both genetically and physical contamination. Nucleus seed should retain original vigour of the variety or parental line. A pedigree certificate is issued by the producing breeder.

Breeder seed:-

Breeder seed is seed or vegetative propagating material directly controlled by the originating or sponsoring plant breeder of the breeding programme or institution and/or seed whose production is personally supervised by a qualified plant breeder and which provides the source for the initial and recurring increase of Foundation seed.

Breeder seed shall be genetically pure (equal or higher than foundation seeds) to guarantee that in the subsequent generation i.e. certified Foundation seed class shall conform to the prescribed standards of genetic purity. The other quality factors of Breeder seed such as physical purity, inert matter, germination etc. shall be indicated on the label on actual basis.

A golden yellow colour certificate is issued for this category of seed by the producing breeder.

Foundation Seed:-

Foundation seed shall be the progeny of Breeder seed, or be produced from Foundation seed which can be clearly traced to Breeder seed. Thus, Foundation seed can even be produced from Foundation seed. During the production of certified Foundation seed, the following guidelines shall be observed:

- (a) Certified Foundation seed produced directly from Breeder seed shall be designated as Foundation seed stage-I;
- (b) Certified Foundation seed produced from Foundation seed stage-I shall be designated as Foundation seed stage-II;
- (c) Certified Foundation seed stage-II will not be used for further increase of Foundation seed and shall be used only for production of Certified seed class;
- (d) Minimum Seed Certification Standards shall be the same for both Foundation seed stage-I and II unless otherwise prescribed;
- (e) **Certification tag shall be of white colour** for both Foundation seed stage-I and II and shall contain the information as to its stage;
- (f) Production of Foundation seed stage-II shall ordinarily be adopted in respect of such crop varieties provided, when it is expressly felt by the Certification Agency that Breeder seed is in short supply;
- (g) Production of Foundation seed stage-II may be adopted for the following group of crops:
 - vegetatively propagated crops;
 - apomictically reproduced crops;
 - self-pollinated crops;
 - often cross-pollinated and cross-pollinated crops, these being gene – pools should not lose their genetic identity and purity if measures to safeguard the same are adequately taken;
 - composite and synthetics;
 - parental line increase of hybrids.

Production of Foundation seed stage-I and II shall be supervised and approved by the Certification Agency and be so handled as to maintain specific genetic identity and genetic purity and shall be required to conform to certification standards specified for the crop/variety being certified.

Certified Seed: -

- (a) Certified seed shall be the progeny of Foundation seed and its production shall be so handled as to maintain specific genetic identity and purity according to standards prescribed for the crop being certified;
- (b) Certified seed may be the progeny of Certified seed provided this reproduction does not exceed three generations beyond Foundation seed stage-I and
 - it is determined by the Certification Agency that genetic identity and genetic purity will not be significantly altered;
 - and when the Certification Agency is satisfied that there is genuine shortage of Foundation seed despite all the reasonable efforts made by the seed producer.
- (c) Certification tag shall be of blue colour (shade AZURE BLUE) for certified seed class.
- (d) Certified seed produced from certified seed shall not be eligible for further seed increase under certification. Certification tags for such production which

is not eligible for further seed increase under certification shall be super scribed with, “not eligible for further seed increase under certification”.

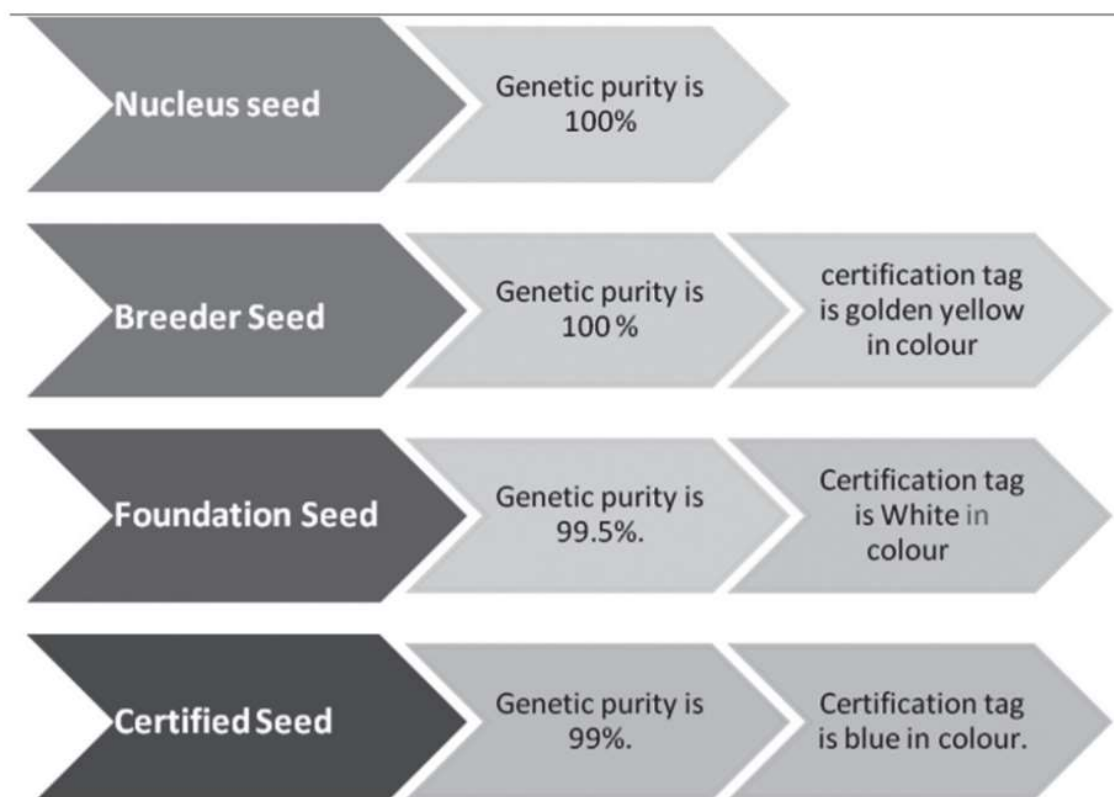
Registered seed: Registered seed shall be the progeny of foundation seed that is so handled as to maintain its genetic identity and purity according to standard specified for the particular crop being certified. A purple colour certificate is issued for this category of seed.

Hybrid seed: The first generation resulting from crossing of two varieties or parents. Hybrid(certified) seed is the first generation resulting from the cross of two approved inbred lines or parents , one of which is male sterile.

Labeled Seed: The seed notified under Section 5 of the Seeds Act, 1966, such seed sold in the market has to be labeled as prescribed under Section 6(a) and (b) of the Seeds Act. Such seed is called Labeled Seed.

The foundation and certified seeds can be multiplied at stage 1 and II, but the reproduction can not exceed three generations after breeder seed.

Certified seed	Truthful labelled seed
Certification is voluntary. Quality guaranteed by certification agency.	Truthful labelling is compulsory for notified kind of varieties. Quality guaranteed by producing agency
Applicable to notified kinds only	Applicable to both notified and released varieties
It should satisfy both minimum field and seed standards	Tested for physical purity and germination
Seed certification officer, seed inspectors can take samples for inspection	Seed inspectors alone can take samples for checking the seed quality.



Seed Multiplication Ratio (SMR): It is nothing but the number of seeds to be produced from a single seed when it is sown and harvested.

SMR = Seed Yield/ Seed Rate

Wheat	1:20
Paddy	1:80 (Varieties)
	1:100 (Hybrids)
Maize	1:80 (Varieties)
	1:100 (Hybrids)
Sorghum	1:100
Bajra	1:200
Blackgram/ greengram	1:40

Seed Replacement Rate (SRR) Seed replacement rate is the percentage of area sown out of total area of crop planted in the season by using certified / quality seeds other than the farm saved seeds.

SRR = $X / Y \times 100$ Where,

X = Quantity of farmer saved seed

Y = Quantity of quality seeds of a particular variety reported to cover a given area.

This is essential for maintaining genetic purity and quality seed production. The seed replacement rate gives an idea about the quantity of the quality seeds used by the farmers.

Seed Replacement and Seed Multiplication Ratio for Different Crops

S. No.	Name of the crop	SRR	SMR
1	Paddy	17	1:80
2	Bajra (Pearl millet)	8	1:200
3	Maize	6	1:80
4	Redgram	6.1	1:100
5	Blackgram	17.7	1:40
6	Greengram	11.7	1:40
7	Cowpea	14.2	1:40
8	Groundnut	5	1:8
9	Sunflower	50	1:50
10	Sesame	15	1:250

Seed Rate for different crops:

Paddy:

Optimum seed rate of 50 kg/ha for upland, 25 kg/ha for lowland and hilly region

Yield: Average yield of rice varieties in case of upland rice is 2.0-2.5 tonnes/ha, lowland rice 4.5-5.0 tonnes/ha, hilly region 3.0-3.5 tonnes/ha. With respect to rice hybrids in irrigated ecology is 5.5-6.0 tonnes/ha.

Wheat:

For medium sized grain varieties Seed rate of 100 kg/ha, for bold seeded variety 125 kg/ha and for late sown seed rate of 125-150 kg/ha is recommended.

Yield: Under irrigated conditions the yield may vary from 2.5 to 6.5 tonnes /ha depending upon practices followed.

Maize: for normal Grain 20 Kg /ha, For Fodder 50 Kg/ha.

Yield: By adopting package of practices as indicated above; it is possible to obtain 50-60 qtl/ha grain in case of hybrid and 40-50 qtl/ha of grain in case of composites.

Sorghum: seed rate of 8-10 kg/ha is recommended to obtain required plant population. Average yield is 8-14 qtl/ha.

Pearl Millet: The seed rate of 5 kg/ha is required to obtain the required crop stand. Yield: Under Irrigated condition, Grain-23-35 qtl/ha and Fodder-100-120 qtl/ha. Under Rainfed condition, Grain-12-15 qtl/ha and Fodder-70-75 qtl/ha.

Barley: Seed rate of 75-80 kg/ha may be used for irrigated timely sown, whereas a seed rate of 100 kg/ha is suitable for irrigated late-sown and saline-alkaline soils. For rainfed areas, a seed rate of 80-100 kg/ha should be used depending on soil-moisture status at the sowing. In the cool arid region of Ladakh a higher seed rate of 250-300 kg/ha may be used.

Yield: The average yield of rainfed crop ranges between 2,000 and 2,500 kg/ha. whereas that of irrigated crop is twice as much.

Blackgram:

Seed rate is 15-20 kg/ha for kharif and 25-30 kg/ha for spring or rabi.

yield: The yield is around 10-12 q/ha.

Greengram:

The seed rate is 25-30 kg/ha for spring or summer and 15-20 kg/ha for kharif and rabi.

Yield: The average yield varies from 6 to 8 q/ha.

Lentil:

The seed rate is 30-40 kg/ha for small –seeded varieties and 50-60 kg/ha for bold seeded varieties.

Yield: The average yield is 8 to 10 q/ha under rainfed and 12-15 q/ha under irrigated condition.

Pea: The seed rate is 70-80 kg/ha.

Yield: The average yield under irrigation is 15-20 q/ha.

Seed producing agencies Seed production is specialised cultivation of a crop under the supervision of trained personnel or experts. The authorised sources of seeds in the country are:

- Indian Council of Agricultural Research (ICAR) institutions
- State Agricultural Universities (SAUs)
- Sponsored breeders recognised by selected State Seed Corporations
- National Seeds Corporation (NSC)
- State Seeds Corporation (SSC)
- State Farms Corporation of India (SFCI)
- Krishi Vigyan Kendras (KVKs)
- Non-governmental Organisations, etc.

National Seeds Corporation Ltd.(NSC), a Public Sector Undertaking under the administrative control of the Department of Agriculture and Cooperation, was established in the year 1963 under the Companies Act, 1956 with the objective of producing and distributing Seeds of high quality to the farmers. The Corporation undertakes the production of seeds through Contract Growers. NSC is dealing with about 560 varieties in 79 crops.

The International Seed Testing Association (ISTA):

Objectives of the Association (a) To develop, adopt and publish standard procedures for sampling and testing seeds, and to promote uniform application of these procedures for evaluation of seeds moving in international trade.

Seed Processing:

- Seed processing is a vital part of the total technology involved in making available high quality seed.
- It assures the end users, seeds of high quality with minimum adulteration.
- In Agriculture, the term seed processing includes cleaning, drying, seed treatment, packaging and storage.
- Seed processing may be understood to 'comprise all the operations after harvest that aim at maximizing seed viability, vigour and health.

Purposes of seed processing:

- To lower the cost of further processes like storage including transport.
- This is achieved by reducing the bulk of the seed lot by cleaning debris and by removing empty or fractured seed (pre- cleaning)
 1. To increase the longevity of seeds; by drying seeds to safe moisture content and treating with protective chemicals
 2. To reduce the variability in vigour by invigorating the seeds and removing the low vigour seeds
 3. To improve the uniformity in seed shape or size by grading or by pelleting.

Sequences of operations are based on characteristics of seed such as shape, size, weight, length, surface structure, colour and moisture content. Because each crop seed possesses individually seed structure.

- Therefore, sequence of operation will be applied using proper equipments.
- However, sequences of operation in seed processing are drying, receiving, pre-cleaning, conditioning, cleaning, separating or upgrading, treating (Drying), weighting, bagging and storage or shipping.

Seed Village Concept: A village, wherein trained group of farmers are involved in production of seeds of various crops and cater to the needs of themselves, fellow farmers of the village and farmers of neighboring villages in appropriate time and at affordable cost is called "a seed village"

Objectives:

- Increasing the seed production
- Increasing the seed replacement rate
- Organizing seed production in cluster (or) compact area replacing existing local varieties with new high yielding varieties
- Self sufficiency and self reliance of the village
- To meet the local demand, timely supply.

The present programme of seed village scheme:

- To upgrade the quality of farmer saved seed financial assistance for distribution of foundation/certified seeds at 50% cost of the seed for production of quality seeds.
- Assistance to train the farmers on seed production and seed technology @ Rs.15000/- for a group of 50-150 farmers.
- Assistance @ 33% subject to a maximum of Rs.3000/- for SC/ST farmers and @ 25% subject to maximum of Rs.2000 for other farmers for procuring seeds storage bin of 20 qtl. capacity.

INTENSIVE FARMING: Intensive Farming refers to an agricultural system, wherein there is high level use of labor and capital, in comparison to the land area. It is practiced in densely populated region.

EXTENSIVE FARMING: Extensive Farming is a farming technique, in which large farms are being cultivated, with relatively lower inputs, i.e. capital and labor. It is practiced in moderately populated region.

Contour farming: the practice of tilling sloped land along lines of consistent elevation in order to conserve rainwater and to reduce soil losses from surface erosion. These objectives are achieved by means of furrows, crop rows, and wheel tracks across slopes, all of which act as reservoirs to catch and retain rainwater, thus permitting increased infiltration and more uniform distribution of the water. Tillage erosion is the soil movement and erosion by tilling a given plot of land. A similar practice is contour bunding where stones are placed around the contours of slopes. Contour ploughing helps to reduce soil erosion.

Apiculture (Beekeeping) The process of rearing of honeybees for the production of honey, bee wax and royal jelly is known as beekeeping. The honeybee rearing is household practice in India. A location where bees are kept is called an apiary or "bee yard". Honeybees and their usefulness are known to man from prehistoric times.

- **Swarming:** Swarming is a natural method of colony multiplication in which a part of the bee colony migrates to a new site to make a new colony. Swarming occurs when a colony builds up a considerable strength or when the queen's substance secreted by queen falls below a certain level. Swarming is a potent instinct in bees for dispersal and perpetuation of the species.

Lac culture Lac is a resinous exudation from the body of female scale insect. Since Vedic period, it has been in use in India. The English word lac synonyms Lakh in Hindi, which itself is derivative of Sanskrit word Laksh meaning a lakh or hundred thousand. It would appear that Vedic people knew that the lac is obtained from numerous insects and must also know the biological and commercial aspects of lac industry. With increasing universal environment awareness, the importance of lac has assumed special relevance in the present age, being an eco-friendly, biodegradable and self-sustaining natural material.

Sericulture Sericulture is the rearing of silkworms for the production of silk. The silkworm larve feed on the mulberry leaves and then they spin their silken cocoons on the twigs. This process is achieved by secretion of fluid from its structural glands, resulting in the fiber of the cocoon. The silk is a continuous filament comprising fibroin protein, secreted from two salivary glands in the head of each larva, and a gum called sericin, which cements the filaments and forms the silk. Bombyx mori is the most widely used silkworm. Sericulture has become an important industrial business in Brazil, China, France and India etc.

Pisciculture Pisciculture is a scientific technology for rearing of fishes for getting maximum fish production from a pond or a tank through utilisation of available food organisms supplemented by artificial feeding. Major species selected for pisciculture are Katla, Rohu, Mrigal, and exotic or common carps. The basic requirements for pisciculture is pond/tank, liming & manuring, fish culture, fish food.

Mushroom culture Mushrooms are most important fungi used as a source of human food. They are a rich source of nutrition. While more than 2,000 species of fungi are edible

but only 5–6 have been commercially exploited the world over. In India, only three mushroom types— white button mushroom (*Agaricus bisporus*), oyster mushroom (*Pleurotus* spp.) and paddy straw mushroom (*Volvariella* spp.) are being cultivated commercially. Button mushroom comprises more than 90% of India's total mushroom production. All the 3 cultivated mushrooms not only differ in their shape, size, colour and biochemical composition, but they show variations in their requirement for growth medium (substrate), temperature, and other physical factors.